Supplier Performance

8 KPIs were measured across 19 suppliers of a memory chip maker.

Each KPI compares 2025 performance against 2024.

- On Time Delivery (OTD)
- On Time In Full (OTIF)
- Lead Time Variability
- Supplier Defect Rate (PPM)
- Purchase Price Variance (PPV)
- Total Cost of Ownership (TCO)
- Cost Savings and Avoidance
- Supplier Scorecard

78% → 86%
On time in full, 2024 to 2025

7 → 3
Suppliers above 1,000 PPM

$11.5M
Savings from 9 sourcing events

$315K
Yearly cost of picking the cheapest NF3 bid

Halvorn Memory Technologies
Excel and Tableau

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On Time Delivery (OTD) and On Time In Full (OTIF)

Delivery was improved in 2025, but 14% of orders were still received late or short.

WHAT IT MEASURES

The percentage of orders received by the promised date (OTD), and the percentage received on time and in full quantity (OTIF).

WHY IT MATTERS

Production can be stopped by a late or short delivery, so the suppliers that can be planned around are identified.



LIMITATION

A short shipment is still counted as on time by OTD, so OTD should be read alongside OTIF.

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Lead Time Variability

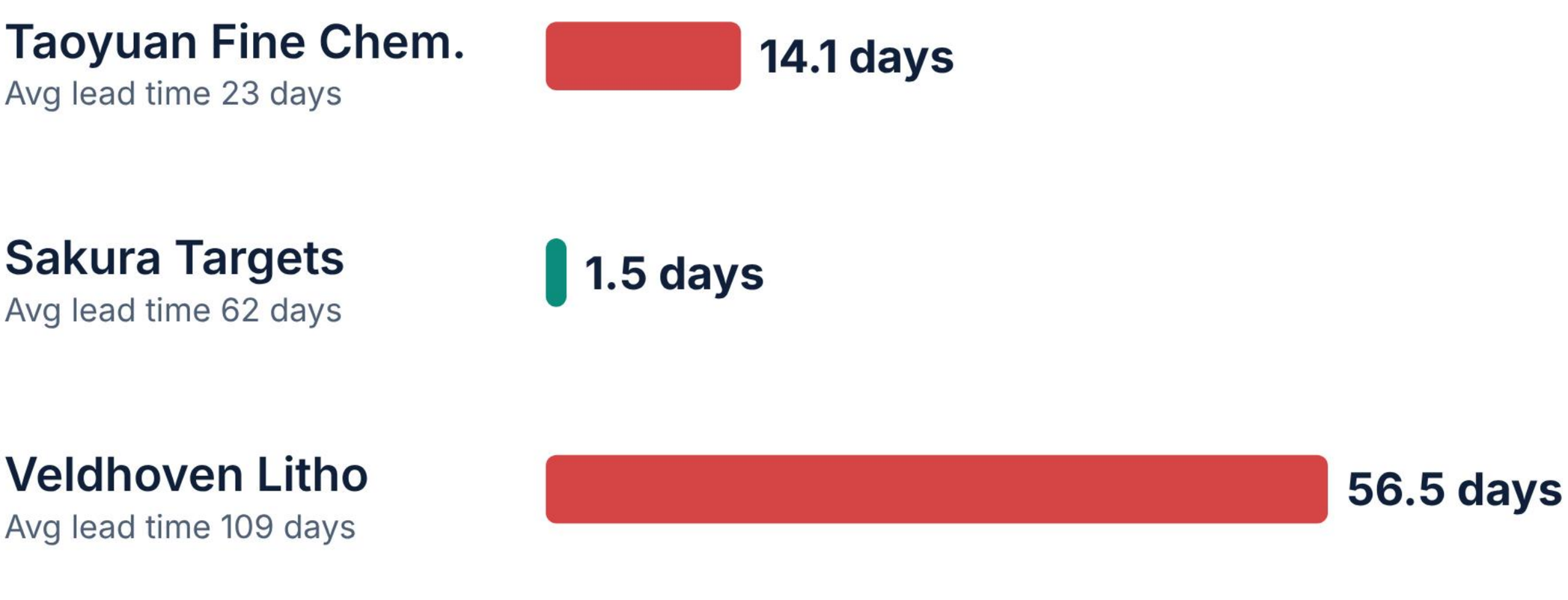
Consistency in delivery time was found to matter more than speed when deciding how much safety stock is held.

WHAT IT MEASURES

How much a supplier's delivery time changes from one order to the next.

WHY IT MATTERS

Extra stock is held when delivery time is unpredictable, and cash is tied up in that stock.



Bars: days of safety stock needed at 95% service level, 2024

LIMITATION

When few orders are placed, one late delivery can make a supplier appear less reliable than it is.

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Supplier Defect Rate, Parts Per Million (PPM)

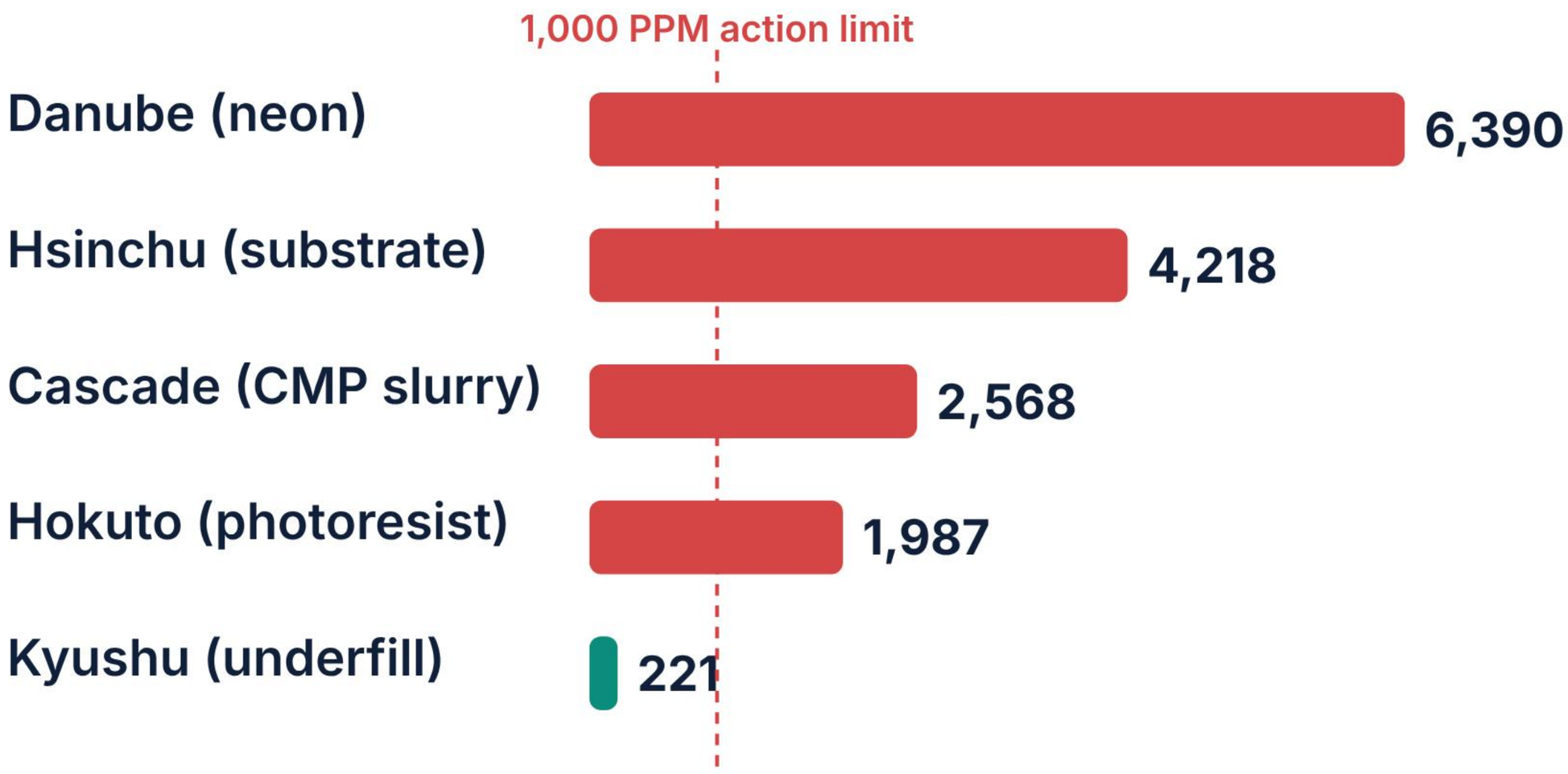
Suppliers above the 1,000 PPM action limit were reduced from **41% to 20%**.

WHAT IT MEASURES

The number of parts rejected for every one million received. 1,000 PPM means 1 bad part in every 1,000.

WHY IT MATTERS

The suppliers causing scrap, rework and line stoppages are identified.



LIMITATION

Only defects caught at inspection are counted. Bad parts that slip through are found later on the line as yield loss.

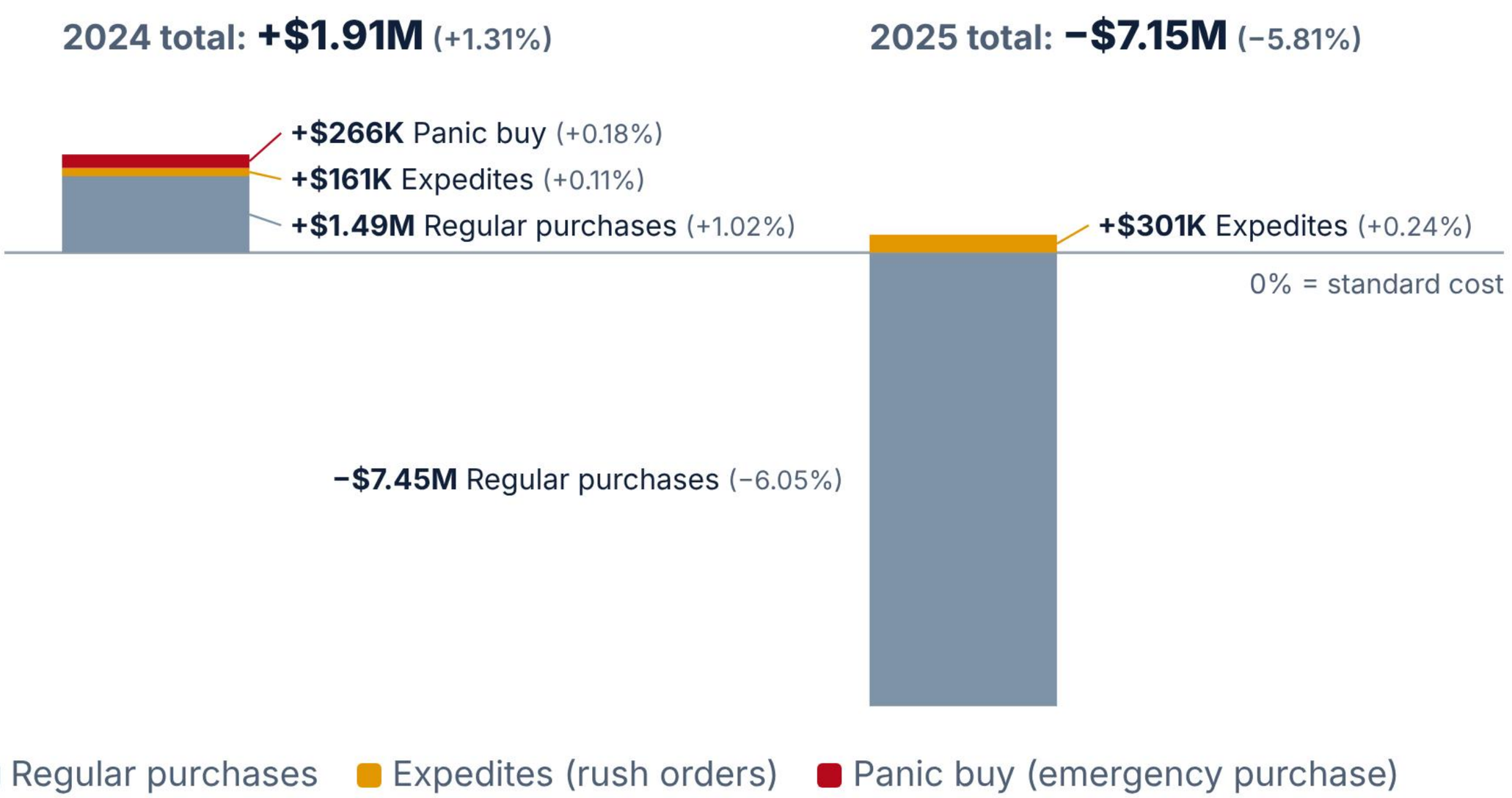
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Purchase Price Variance (PPV)

Purchase prices were moved from 1.3% above standard cost in 2024 to 5.8% below in 2025.

WHAT IT MEASURES
The difference between the price paid and the planned standard cost.

WHY IT MATTERS
Overpayment, rush shipping and panic buying are exposed.



LIMITATION
A favorable variance can be caused by an outdated standard cost rather than a better deal.

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Total Cost of Ownership (TCO)

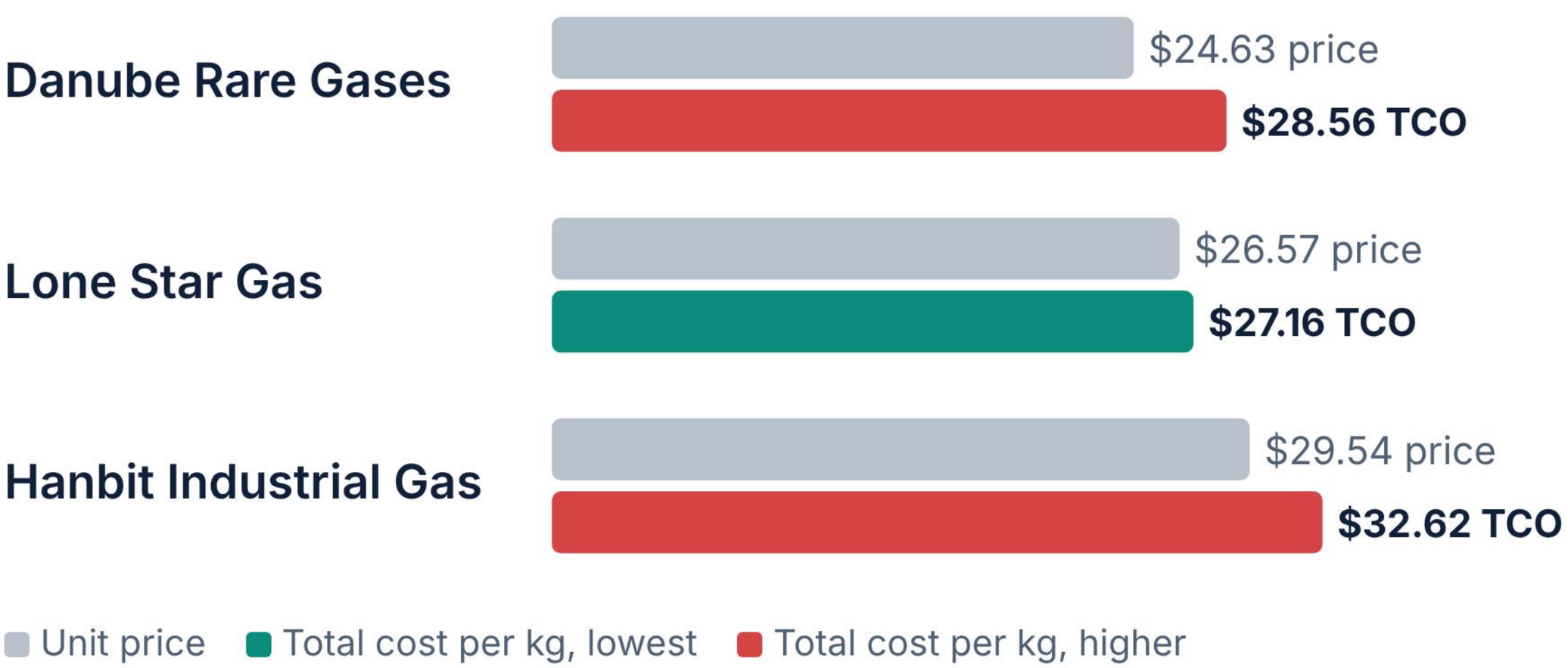
The cheapest NF3 bid cost $1.40/kg more on total cost, about $315K a year.

WHAT IT MEASURES

Price plus freight, defect cost, delivery risk, payment terms and qualification cost.

WHY IT MATTERS

The supplier that is truly cheapest is identified before a contract is awarded.



LIMITATION

The TCO ranking depends on assumed costs for defects, late delivery and payment terms. If those assumptions are changed, a different supplier can come out cheapest.

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Cost Savings and Cost Avoidance

$11.5M was saved across 9 sourcing events, and $1.05M was avoided in 4 negotiations.

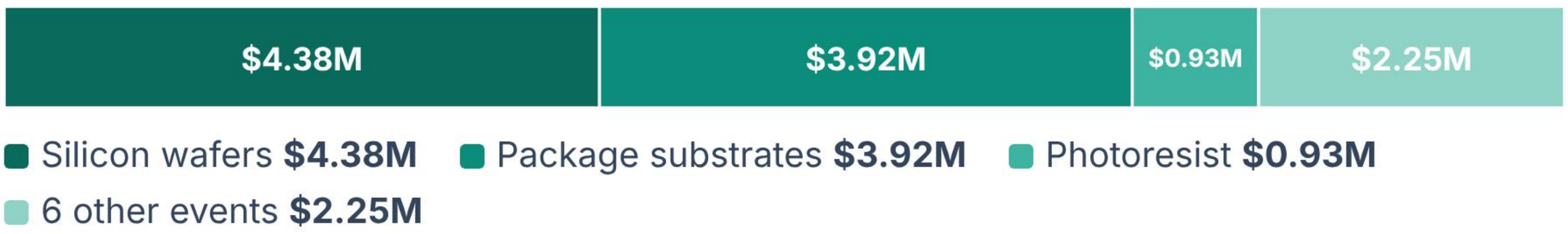
WHAT IT MEASURES

Savings are counted when a lower price is negotiated. Avoidance is counted when the impact of a price increase is reduced or a new expense is prevented entirely.

WHY IT MATTERS

The money saved, and the extra cost stopped through negotiation, are both shown in dollars.

Cost savings, 9 sourcing events — $11.48M



Cost avoidance, 4 negotiations — $1.05M



LIMITATION

Avoided cost is money that was never spent, so it is not seen in the budget. Only savings lower the actual cost, which is why the two are kept separate.

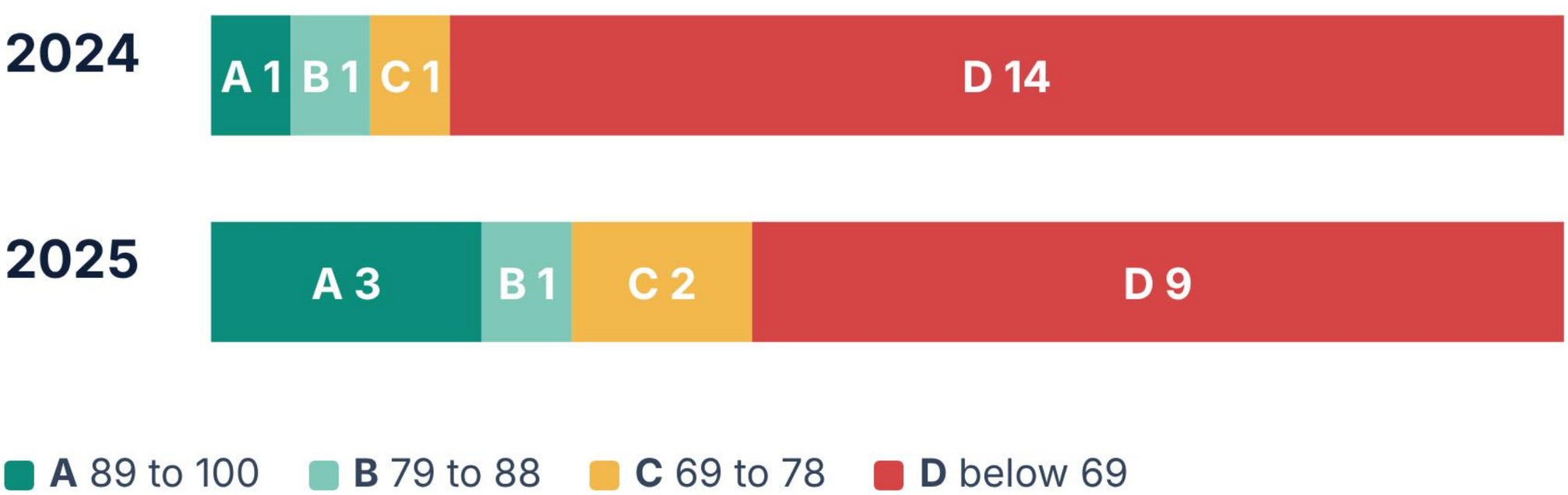
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Supplier Scorecard

With industry benchmark bands, 12% of suppliers were graded A or B in 2024 and 27% in 2025.

WHAT IT MEASURES
One weighted score combining delivery, quality, cost and lead time reliability.

WHY IT MATTERS
Decides which suppliers to grow, develop or replace.



SCORE WEIGHTS

35%	35%	20%	10%
Delivery (OTIF)	Quality (PPM)	Cost (PPV)	Lead time variability

LIMITATION
Grades depend on the weights and score limits chosen. If they are changed, the same supplier can receive a different grade.

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Decisions From the Data

1. Issue corrective actions to the 3 suppliers above 1,000 PPM, starting with neon.
2. Award contracts on total cost of ownership, not unit price.
3. Review OTIF and lead time variability together in every supplier review.
4. Keep the second photoresist source, even though it was paid 3.1% above standard cost.

Explore it yourself

Interactive Tableau dashboard and the Excel workbooks with every formula are linked in the post.

Next week: Operations. Equipment effectiveness, yield, cycle time and unplanned downtime.

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